



اَبْنُو سَيِّدِي تَكُونُ لِي مَارَا
UNIVERSITI
TEKNOLOGI
MARARA

ITT300

INTRODUCTION TO DATA COMMUNICATION AND NETWORKING

PROJECT PACKET TRACER REPORT

CLASS: M3CS1104C

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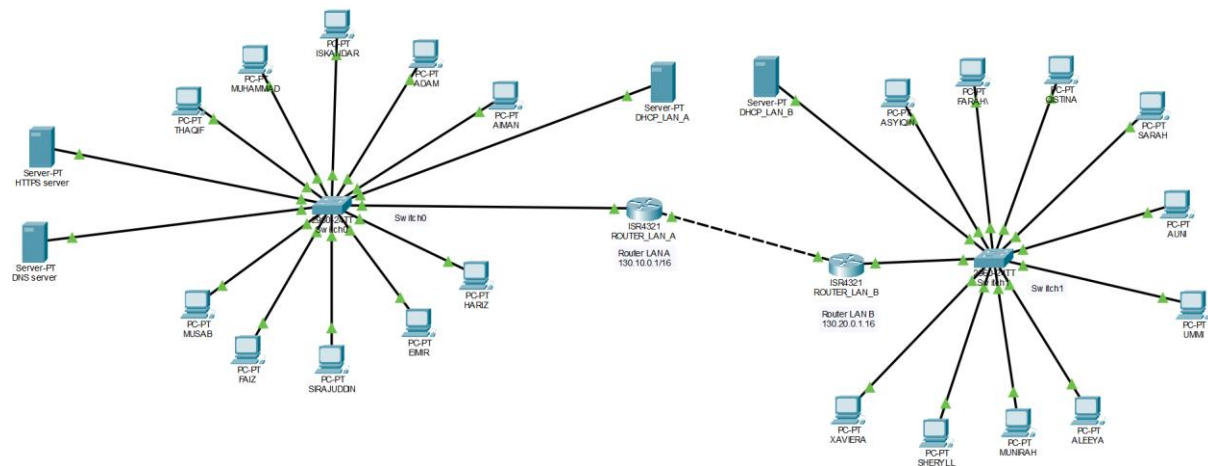
INTRODUCTION

This report documents a group project completed for ITT300: Introduction to Data Communication and Networking, in which Cisco Packet Tracer was used to design, configure, and test a small enterprise network. The objective of the project was to apply core networking concepts covered in the course, including network topology design, IP addressing, static routing between routers, DHCP-based address allocation, and DNS/HTTP service configuration, within a simulated environment.

The network consists of two Local Area Networks, LAN A and LAN B, each built around a switch connecting ten PCs and linked to the wider network through a router. The two routers are connected to each other to enable inter-subnet communication. Task 1 covers the overall network design, Task 2 covers the construction and verification of the two LAN topologies, Task 3 covers router IP addressing and inter-router connectivity, Task 4 covers the configuration of DHCP services for automatic IP address assignment, and Task 5 covers the configuration of DNS and HTTP services to host and access a web page by domain name. Screenshots from Packet Tracer are provided throughout this report as evidence of successful configuration and testing at each stage.

TASK 1

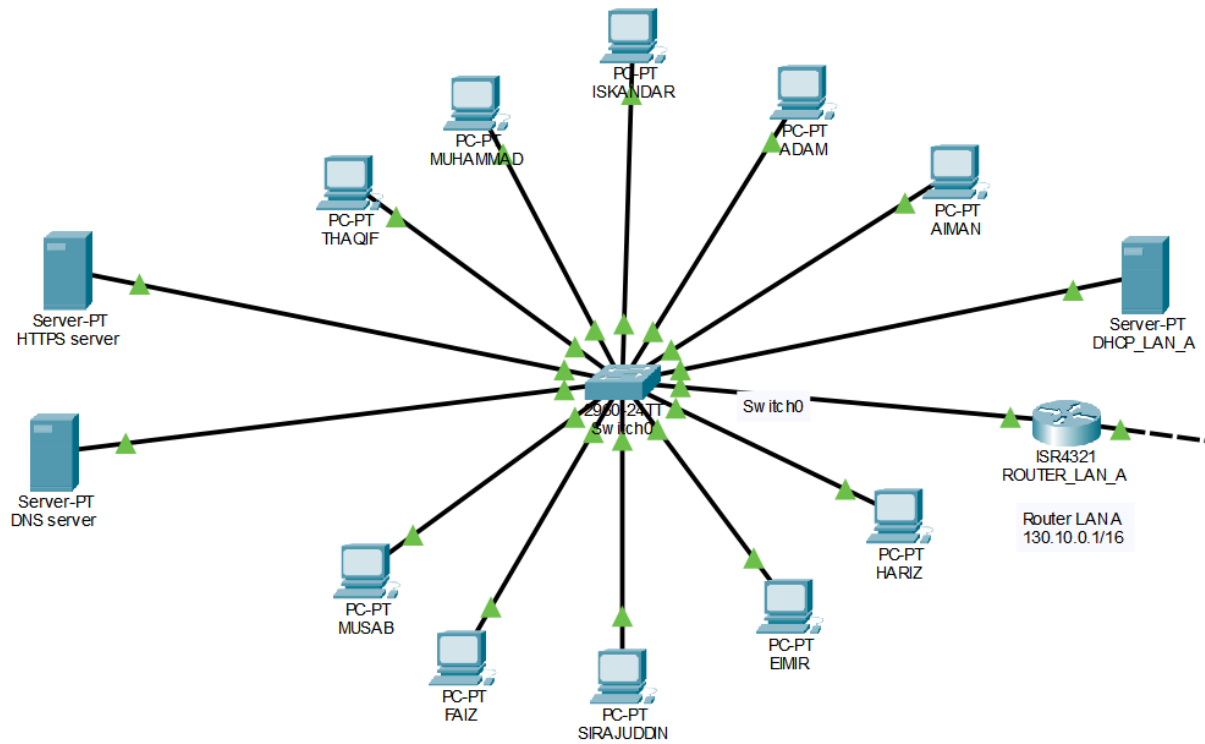
One network design with **10 Personal Computers for each switch, 2 Switches and 2 Routers.**



This diagram shows the completed network design for Task 1, consisting of two routers, two switches, and ten PCs connected to each switch.

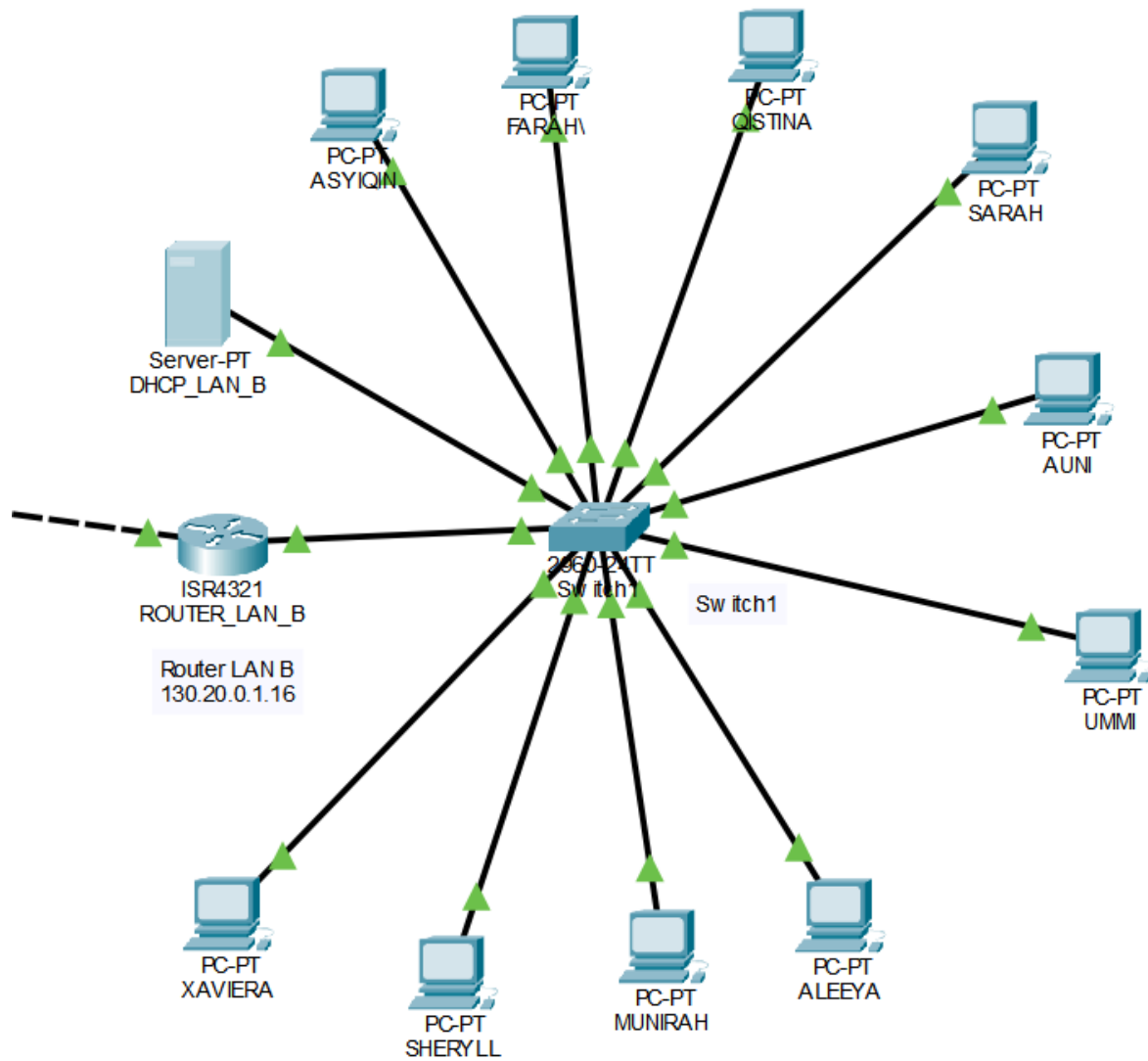
TASK 2

LAN A Topology



This topology shows how the PCs, switch, and router are interconnected within LAN A.

LAN B Topology



This topology shows the equivalent network arrangement built for LAN B.

Successful pinging from LAN A to LAN B

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 130.20.0.100/16
Ping request could not find host 130.20.0.100/16. Please check the name and try again.
C:\>ping 130.20.0.100

Pinging 130.20.0.100 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126

Ping statistics for 130.20.0.100:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

The successful ping results confirm that devices in LAN A can communicate with devices in LAN B.

LAN A PC ipconfig

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: FE80::20D:BDFF:FE12:DEA4
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 130.10.0.104
    Subnet Mask . . . . .: 255.255.0.0
    Default Gateway . . . . .: ::
                                   130.10.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . .: ::
    IPv4 Address. . . . .: 0.0.0.0
    Subnet Mask . . . . .: 0.0.0.0
    Default Gateway . . . . .: ::
                                   0.0.0.0
```

The ipconfig output verifies the IP address, subnet mask, and default gateway assigned to a PC in LAN A.

LAN B PC ipconfig

```
C:\>ipconfig

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: FE80::204:9AFF:FE7D:178E
    IPv6 Address . . . . . : ::
    IPv4 Address. . . . . : 130.20.0.100
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : ::
                               130.20.0.1

Bluetooth Connection:

    Connection-specific DNS Suffix...: 
    Link-local IPv6 Address . . . . .: ::
    IPv6 Address . . . . . : ::
    IPv4 Address. . . . . : 0.0.0.0
    Subnet Mask . . . . . : 0.0.0.0
    Default Gateway . . . . . : ::
                               0.0.0.0
```

The ipconfig output verifies the IP configuration assigned to a PC in LAN B.

TASK 3

Configured the right IP address for router 0

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     130.10.0.1      YES manual up          up
GigabitEthernet0/0/1     130.30.0.1      YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
```

This configuration confirms that Router 0 has been assigned the correct IP address on its interface.

Configured the right IP address for router 1

```
Router>show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0/0     130.20.0.1      YES manual up          up
GigabitEthernet0/0/1     130.30.0.2      YES manual up          up
Vlan1                    unassigned      YES unset  administratively down down
```

This configuration confirms that Router 1 has been assigned the correct IP address on its interface.

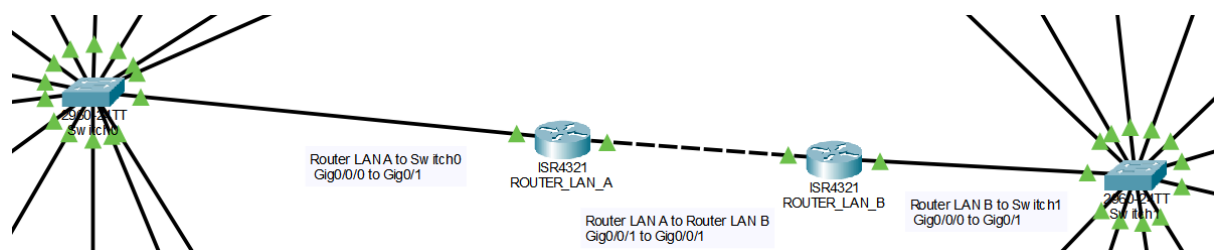
Pinging router 0 from router 1

```
Router>ping 130.10.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.10.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

The ping result verifies that Router 1 can successfully reach Router 0.

Two routers connected with each other



This shows the two routers connected to each other, forming the link between LAN A and LAN B.

TASK 4

DHCP for LAN A

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask
serverPool	130.10.0.1	130.10.0.21	130.10.0.100	255.255.0.0

This configuration sets up the DHCP pool that automatically assigns IP addresses to devices in LAN A.

DHCP for LAN B

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask
serverPool	130.20.0.1	130.10.0.21	130.20.0.100	255.255.0.0

This configuration sets up the DHCP pool that automatically assigns IP addresses to devices in LAN B.

A PC from LAN A automatically pulled an IP address from the GUI pool

```
FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::20D:BDFE:FE12:DEA4
IPv6 Address.....: ::
IPv4 Address.....: 130.10.0.104
Subnet Mask.....: 255.255.0.0
Default Gateway.....: ::
                        130.10.0.1
```

This confirms that a PC in LAN A successfully obtained an IP address automatically from the DHCP pool.

A PC from LAN B automatically pulled an IP address from the GUI pool

```
FastEthernet0 Connection:(default port)

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....: FE80::204:9AFF:FE7D:178E
IPv6 Address.....: ::
IPv4 Address.....: 130.20.0.100
Subnet Mask.....: 255.255.0.0
Default Gateway.....: ::
                        130.20.0.1
```

This confirms that a PC in LAN B successfully obtained an IP address automatically from the DHCP pool.

Inter-subnet ping proof

```
C:\>ping 130.20.0.100

Pinging 130.20.0.100 with 32 bytes of data:

Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126
Reply from 130.20.0.100: bytes=32 time<1ms TTL=126

Ping statistics for 130.20.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

This ping result proves that devices on different subnets can communicate with one another through the routers.

TASK 5

DNS server configuration

Name		Type	A Record
Address			
No.	Name	Type	Detail
0	80300dec2025.ultn.edu.my	A Record	130.10.0.20

This shows the DNS server configuration used to resolve the domain name to its IP address.

HTTP server edit

HTTP

☒ On
 ☐ Off

HTTPS

☒ On
 ☐ Off

File Manager

	File Name	Edit	Delete
1	Picture1.jpg		(delete)
2	Picture2.jpg		(delete)
3	Picture3.jpg		(delete)
4	adam.jpg		(delete)
5	copyrights.html	(edit)	(delete)
6	cscopptlogo177x111.jpg		(delete)
7	helloworld.html	(edit)	(delete)
8	image.html	(edit)	(delete)
9	index.html	(edit)	(delete)
10	ultn.jpg		(delete)

This shows the HTTP server settings being edited to host the web page.

Ping by domain name

```
C:\>ping itt300dec2025.uitm.edu.my

Pinging 130.10.0.20 with 32 bytes of data:

Request timed out.
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126

Ping statistics for 130.10.0.20:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

This ping result confirms that the domain name successfully resolves to its corresponding IP address.

Ping by IP address

```
C:\>ping 130.10.0.20

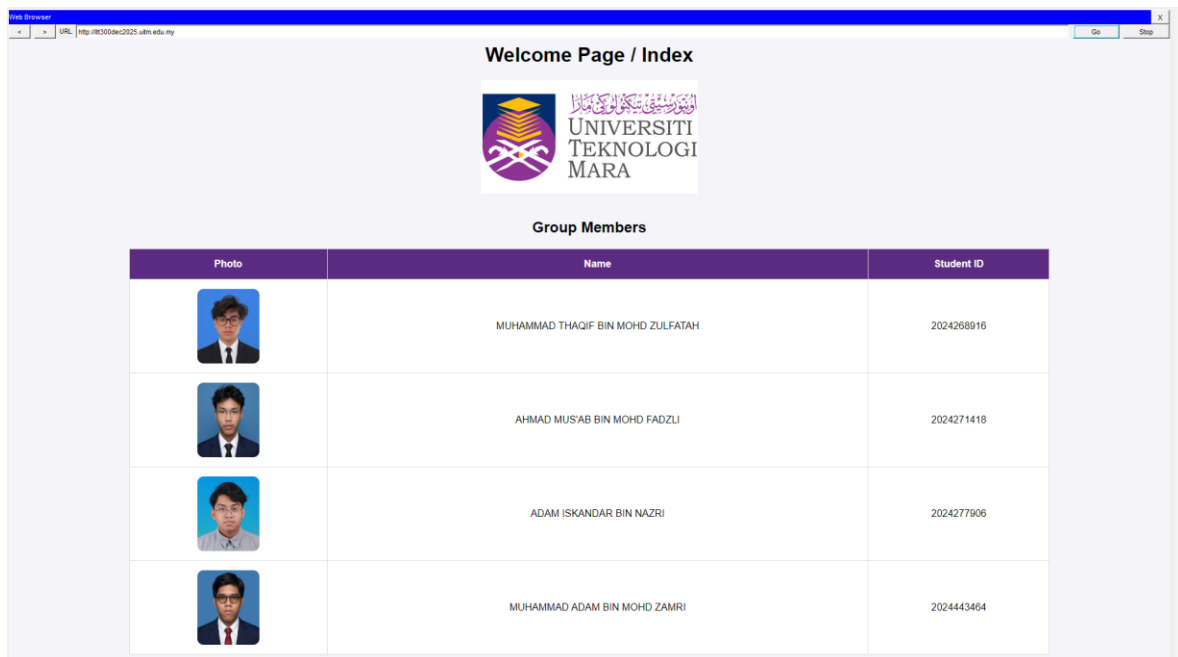
Pinging 130.10.0.20 with 32 bytes of data:

Reply from 130.10.0.20: bytes=32 time<1ms TTL=126
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126
Reply from 130.10.0.20: bytes=32 time<1ms TTL=126

Ping statistics for 130.10.0.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

This ping result confirms connectivity to the server using its IP address directly.

index.html



This shows the web page as displayed in the browser.

index.html coding

File Name: index.html

```
<html>
<head>
<title>Welcome Page / Index</title>
<style>
  body { font-family: Arial, sans-serif; text-align: center; background-color: #f4f4f9; }
  .logo { width: 350px; margin-bottom: 20px; }
  .student-photo { width: 100px; height: 120px; border-radius: 10px; }
  table { margin: 0 auto; background-color: white; border-collapse: collapse; width: 80%; }
  th, td { border: 1px solid #ddd; padding: 15px; text-align: center; font-size: 16px; }
  th { background-color: #5b2b82; color: white; }
</style>
</head>
<body>

<h1>Welcome Page / Index</h1>



<h2>Group Members</h2>
<table>
  <tr>
    <th>Photo</th>
    <th>Name</th>
    <th>Student ID</th>
  </tr>
  <tr>
    <td></td>
    <td>MUHAMMAD THAQIF BIN MOHD ZULFATAH</td>
    <td>2024268916</td>
  </tr>
  <tr>
    <td></td>
    <td>AHMAD MUS'AB BIN MOHD FADZLI</td>
    <td>2024271418</td>
  </tr>
  <tr>
    <td></td>
    <td>ADAM ISKANDAR BIN NAZRI</td>
    <td>2024277906</td>
  </tr>
  <tr>
    <td></td>
    <td>MUHAMMAD ADAM BIN MOHD ZAMRI</td>
    <td>2024443464</td>
  </tr>
</table>

</body>
</html>
```

This shows the HTML code used to build the web page.

LINK PACKET TRACER

<https://youtu.be/hzXkPtbWfGc>